

KHOMUTOVO, Russian FederationWMO: **321500**Lat: **46.889N**Lon: **142.718E**Elev: **18**StdP: **101.11**Time Zone: **11.00 (E11)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-23.8	-28.8	0.3	-26.2	-25.9	0.4	-23.7	17.5	-6.4	13.5	-6.3	0.9	350	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	26.0	20.2	24.2	19.3	22.8	18.5	21.6	24.3	20.5	22.9	19.5	21.7	3.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.5	23.1	19.8	14.5	22.0	18.7	13.6	20.9	63.1	24.5	59.3	22.8	55.6	21.6	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	8.8	7.8	-30.7	29.5	2.4	1.7	-32.4	30.8	-33.8	31.8	-35.1	32.8	-36.8	34.0		
			-30.9	23.6	2.3	1.3	-32.5	24.5	-33.9	25.3	-35.2	26.0	-36.9	26.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-12.5	-12.0	-5.8	1.8	7.9	12.3	16.4	17.6	13.6	6.5	-1.4	-9.2	(6)
	DBStd	11.28	5.17	5.04	4.52	3.41	3.55	3.22	2.92	2.90	3.29	3.43	4.73	4.71	(7)
	HDD10.0	3209	698	616	489	247	86	14	0	0	7	117	341	596	(8)
	HDD18.3	5635	956	849	748	497	324	182	74	49	146	366	591	854	(9)
	CDD10.0	662	0	0	0	0	20	83	198	237	113	9	1	0	(10)
	CDD18.3	46	0	0	0	0	0	1	14	28	3	0	0	0	(11)
	CDH23.3	251	0	0	0	0	5	17	69	139	21	0	0	0	(12)
	CDH26.7	29	0	0	0	0	1	1	6	20	1	0	0	0	(13)
(14) Wind	WSAvg	3.1	3.3	3.1	3.4	3.5	3.7	3.3	2.9	2.6	2.6	3.0	2.9	3.0	(14)
(15) Precipitation	PrecAvg	880	53	41	44	59	67	59	90	104	103	98	77	66	(15)
	PrecMax	1248	138	106	110	140	134	176	229	319	223	217	142	149	(16)
	PrecMin	656	6	0	7	13	10	0	16	10	10	31	8	13	(17)
	PrecStd	147	29	25	23	29	32	36	50	61	50	46	32	34	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	3.1	6.2	16.1	22.7	25.5	27.3	29.0	25.9	19.2	13.1	4.8	(19)
		MCWB	-0.1	0.7	2.3	8.8	13.7	18.0	20.9	21.5	19.1	13.5	9.1	3.8	(20)
	2%	DB	-0.8	0.8	4.0	13.0	19.8	22.9	25.2	26.7	23.2	16.9	10.2	2.1	(21)
		MCWB	-1.5	-0.5	1.1	7.4	12.5	16.8	20.1	21.0	17.7	12.1	7.6	1.2	(22)
	5%	DB	-2.1	-1.1	2.2	10.1	17.2	20.9	23.3	24.9	21.4	14.9	8.0	0.4	(23)
		MCWB	-2.8	-2.3	0.0	5.8	11.1	15.7	19.1	20.3	17.0	11.3	6.5	-0.4	(24)
	10%	DB	-3.8	-3.0	1.1	7.8	15.0	18.9	21.8	23.0	19.9	13.2	6.0	-1.0	(25)
		MCWB	-4.6	-4.2	-0.8	4.3	9.9	14.6	18.5	19.6	16.2	10.3	4.2	-1.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	1.3	3.2	10.2	14.7	19.4	22.4	23.4	20.6	15.1	10.2	3.7	(27)
		MCDB	0.4	2.3	4.8	15.5	20.7	23.6	25.7	27.1	23.5	17.3	11.7	4.2	(28)
	2%	WB	-1.3	-0.6	1.6	7.7	13.2	17.5	21.1	22.1	19.2	13.5	8.3	1.4	(29)
		MCDB	-0.8	0.5	3.2	11.8	18.3	21.7	24.1	25.0	21.6	15.5	9.6	1.8	(30)
	5%	WB	-2.9	-2.4	0.5	6.0	11.8	16.1	19.9	21.1	18.0	12.2	6.3	-0.2	(31)
		MCDB	-2.4	-1.4	1.9	9.6	16.3	19.9	22.3	23.4	20.5	14.2	7.7	0.3	(32)
	10%	WB	-4.6	-4.2	-0.5	4.5	10.4	15.0	18.8	20.2	16.9	10.9	4.3	-1.9	(33)
		MCDB	-3.7	-3.2	0.7	7.2	14.2	18.1	21.0	22.3	19.1	12.8	5.8	-1.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	13.5	11.0	9.4	10.8	9.8	8.5	8.8	11.3	11.2	9.2	10.6	(35)
		MCDBR	9.3	13.5	11.1	14.1	15.7	14.5	11.7	12.1	13.3	13.8	12.1	9.9	(36)
		MCWBR	8.9	11.8	8.7	8.9	8.9	8.7	7.1	7.1	8.6	9.8	9.8	9.3	(37)
		MCDBR	8.3	12.4	9.8	13.1	14.3	12.6	10.0	9.7	11.1	11.8	11.1	9.2	(38)
	5% WB	MCWBR	8.1	11.1	8.2	8.6	8.6	8.0	6.5	6.3	7.5	9.1	9.6	9.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.313	0.355	0.415	0.492	0.520	0.494	0.462	0.440	0.397	0.403	0.374	0.336	(40)
	taud		2.045	1.898	1.823	1.810	1.872	2.046	2.213	2.287	2.340	2.152	2.109	2.045	(41)
	Ebn at Noon		731	769	780	760	757	782	804	807	810	722	653	644	(42)
	Edh at Noon		97	141	178	198	193	163	137	122	106	109	93	88	(43)
(44) All-Sky Solar Radiation	RadAvg		1.26	2.11	3.24	4.52	4.85	4.77	4.43	4.10	3.69	2.48	1.39	0.99	(44)
	RadStd		0.14	0.21	0.31	0.47	0.44	0.55	0.54	0.36	0.38	0.15	0.12	0.08	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page